HTML COMPLETE PRACTICAL ASSIGNMENT

Student Learning Portal | Beginner to Advanced HTML5

Project	Build a multi-page Student Learning Portal
Technology	HTML5 with basic CSS only. JavaScript may be used only where an interactive HTML feature requires it.
Pages	Home, About, Courses, Text & Formatting, Lists, Tables, Registration, Gallery, Media, Blog and Contact.
Submission	All source files, assets folder, screenshots and a short explanation of selected code.
Suggested time	7 to 10 days

Your goal: Recreate the supplied multi-page layout reference. Focus on correct HTML structure, meaningful content, attributes and working browser behaviour. Keep CSS simple: readable fonts, spacing, borders, colours and basic layout only.

Project folder and required pages

- **index.html** - Home page and navigation.
- **about.html** - Institute information and semantic layout.
- **courses.html** - Course cards, headings, lists and links.
- **tags.html** - Text formatting, quotations, code and preformatted text.
- **lists.html** - Ordered, unordered, nested and description lists.
- **tables.html** - Marksheets and timetable using rowspan and colspan.
- **registration.html** - Complete student registration form.
- **gallery.html** - Images, responsive pictures, captions and image map.
- **media.html** - Audio, video, subtitles, iframe, PDF, SVG and canvas.
- **blog.html** - Semantic article, aside, FAQ, dialog and popover examples.
- **contact.html** - Address, contact links and accessible contact form.

Every page must include

- HTML5 doctype, html element with lang, head and body.
- A unique page title, UTF-8 charset, viewport and useful meta description.
- A consistent navigation menu linking all pages, plus a meaningful footer.
- Correct heading order, descriptive links and valid nesting.
- A favicon and external stylesheet link. Add a small amount of basic CSS only.

1. Text, links, lists and document structure

Text and formatting: h1-h6, p, br, hr, strong, b, em, i, u, mark, small, del, ins, s, sub, sup, code, kbd, samp, var, pre, blockquote, q, cite, abbr, dfn, address, bdi, bdo, wbr, ruby, rt and rp.

Links and navigation: a, nav and link. Demonstrate internal anchors, links to other pages, external links, mailto, tel, downloadable files, target, rel, title and fragment identifiers.

Lists: ul, ol, li, dl, dt, dd and menu. Include nested lists and demonstrate ordered-list type, start and reversed attributes.

Document structure: html, head, title, meta, base, link, style, script and noscript. Explain the purpose of each in your own words.

2. Tables: required rowspan and colspan practice

Create a student marksheet and a weekly timetable. Your marksheet must have grouped column headings and a total row. Your timetable must merge a day cell across multiple time slots.

Element / attribute	What you must demonstrate
table, caption	A clearly titled data table.
thead, tbody, tfoot	Separate headings, data and summary.
tr, th, td	Correct table rows, header cells and data cells.
colgroup, col	Apply a column-level style or span.
colspan	Merge at least two adjacent columns.
rowspan	Merge at least two rows.
scope, headers, id	Associate table headers accessibly.

3. Forms: all standard input types

Create a registration form with personal details, academic details, preferences, uploads and consent. Include labels, fieldsets, legends, helpful instructions and browser validation.

Input type	Example field
text, password, email, number	Name, password, email, age
tel, url, search	Phone, portfolio, course search
date, time, datetime-local	Birth date, preferred time, appointment
month, week	Joining month, preferred week
color, range	Favourite colour, skill level
file	Profile photo and PDF resume
checkbox, radio	Hobbies and course mode
hidden	Non-visible demo student reference
submit, reset, button, image	Submit, clear, custom action, image submit

Form elements: form, label, input, textarea, button, select, option, optgroup, fieldset, legend, datalist and output. Also demonstrate progress and meter where appropriate.

Form attributes: action, method, enctype, target, autocomplete, novalidate, name, accept-charset, plus input attributes such as required, placeholder, value, checked, selected, multiple, readonly, disabled, min, max, step, minlength, maxlength, pattern, accept, list, autofocus, inputmode, capture and dirname.

Form behaviour: Show GET and POST examples. Use required, type, min/max, minlength/maxlength and pattern validation. Explain that real applications still need server-side validation.

4. Images, SEO, accessibility and tooltips

Images: img, picture, source, figure, figcaption, map and area. Add at least five images, meaningful alt text, width and height, loading="lazy" below the fold, and a responsive picture using srcset and sizes. Create an image map with clickable regions.

SEO metadata: unique title, meta description, charset, viewport, author, robots, canonical link, favicon and Open Graph metadata. Meta keywords may be shown as a historical/demo example, but are not a modern Google ranking requirement.

Tooltip exercise: Add the global title attribute to a button, link, image and abbreviation. Hover or focus the element to see the browser's native tooltip. Explain that title tooltips are not consistently accessible on touch devices and should not contain essential information. Use visible help text and aria-describedby for important instructions.

Accessibility: label every form control; use alt text, meaningful link text, logical headings, keyboard-accessible controls, table headers, aria-label only when a visible label is not suitable, aria-describedby for help text, and appropriate roles only when native HTML is insufficient.

5. Semantic, interactive and advanced HTML

Feature	Required practice
Semantic layout	header, footer, main, nav, section, article, aside, address
Time and data	time with datetime; data with value
Expandable content	details and summary for an FAQ
Dialog	dialog with open and close controls; explain modal behaviour
Popover / tooltip-like UI	popover attribute, popovertarget and popovertargetaction on a button; create a non-modal popover
Progress and measurement	progress and meter with suitable values and labels
Reusable markup	template; explain slot through a simple Web Component demonstration if ready
Script fallback	noscript message when JavaScript is disabled
Custom data	data-* attribute and read it with JavaScript dataset
Editable and draggable	contenteditable, draggable and spellcheck demonstration
Keyboard and language	tabindex, accesskey, lang, dir, translate

Important: A native tooltip can be demonstrated with title, but it is not a complete custom tooltip component. The popover API is a separate feature. Students should show both and explain the difference.

6. Audio, video, embedded content and graphics

- audio and video with controls, source, type, poster, preload, muted and loop.
- track with a captions or subtitles WebVTT file.
- iframe with title, loading and a suitable sandbox demonstration.
- object and embed for a sample PDF or supported embedded resource.
- svg with basic shapes and accessible title/description where appropriate.
- canvas with a small drawing made using JavaScript.

7. Global attributes and browser behaviour

Attribute	Student demonstration
id, class, style, title	Unique target, reusable styling, inline style and native tooltip.
hidden, lang, dir	Hide a non-essential demo panel; language and text direction.
tabindex, accesskey	Keyboard navigation and shortcut; use tabindex carefully.
contenteditable, spellcheck	Editable paragraph with spell checking.
draggable	Drag a labelled item; explain that drag-and-drop needs event handling.
translate	Mark a product name as not to be translated.
data-*	Store a course ID and read it using element.dataset.
role, aria-label, aria-describedby	Accessible naming and help text; prefer native semantics.
inert	Temporarily make a section non-interactive.
popover, popovertarget, popovertargetaction	Show and hide a non-modal popover.
autocapitalize, inputmode, enterkeyhint	Demonstrate mobile input hints where supported.

8. Character entities and special content

- Display literal <, >, &, quotation marks, non-breaking spaces and copyright/trademark symbols using named or numeric character references.
- Show code samples safely using pre and code.
- Demonstrate a right-to-left text sample with dir and bdo, and mixed-direction text with bdi.

9. Required understanding: tag, element, attribute, value and DOM property

For at least 10 examples from your own project, identify the tag name, complete element, each attribute, its value, and (where relevant) the corresponding JavaScript DOM property. Example: in input type="email" id="studentEmail", type and id are attributes; email and studentEmail are their values. In JavaScript, input.type and input.id are DOM properties. The live input.value property may change as the user types.

10. What not to use in a modern project

Recognise obsolete elements such as font, center, marquee, big, strike, frameset, frame, acronym, applet, dir, tt and keygen. Do not use them in the final project. Explain a modern alternative where applicable.

11. Submission and marking rubric

Assessment area	Marks
Document structure, navigation and SEO	10
Text, links, lists and semantic HTML	10
Tables including rowspan and colspan	10
Forms, all 22 input types and validation	20
Images, responsive images, alt text and tooltip examples	10
Media, iframe, SVG and canvas	10
Interactive features: details, dialog and popover	10

Assessment area	Marks
Global attributes, accessibility and entities	10
Code quality, valid nesting and student explanation	10
TOTAL	100

Before submission, validate every HTML page using the W3C Nu HTML Checker: https://validator.w3.org/nu/ . Submit your project folder, screenshots of each page, and be ready to explain your implementation.

Teacher verification questions

- Explain the difference between a tag and an element.
- Show where you used colspan and rowspan and explain the difference.
- Demonstrate every input type and explain at least five validation attributes.
- Show a native title tooltip and a popover. Explain how they differ.
- Explain alt text, lazy loading, responsive images and SEO metadata.
- Identify five semantic elements and explain why you selected them.
- Show one HTML attribute and its related DOM property in browser developer tools.